TAN KAI SHENG

APU STUDENT

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PROFILE

I am a motivated **IoT undergraduate** with strong **hands-on skills** in building electronic systems, sensor integration, cloud connectivity, and full-stack IoT pipelines. **Experienced** with **ESP32, MQTT, Node-RED, ThingSpeak/HiveMQ Cloud, MongoDB**, and hardware sensors. Passionate about prototyping, troubleshooting, and developing real-world IoT solutions. **Seeking an internship (Jan–Apr 2026)** to apply technical skills in a professional engineering environment.

EDUCATION

2024-Present
ASIA PACIFIC UNIVERSITY (APU)

- Bachelor of Science (Honours) in Information Technology with a specialism in Internet of Things (IoT)

2018-2023
KWANG HUA PRIVATE SCHOOL

PROJECTS

2025

Smart Weather Monitoring System

- Created an ESP32-based station with scalability that transmits real-time data of various environmental sensors (DHT11, MQ135, Rain, LDR, UV, BMP180) over HiveMQ Cloud (MQTT).
- Designed a Node-RED dashboard to provide live visualization and MongoDB to log historical data.

Smart Automated Watering System

- Constructed an accurate irrigation system with ESP32, DHT11, LDR (light intensity), and manual override touch sensor.
- Created a bespoke web interface (HTML/CSS) that allows real-time control and monitoring of the SG90 water valve remotely.
- Thing Speak built in to provide real-time data visualization and manual override logic of irrigation by touch.

APU Chinese Language & Culture Society (Events)

APU Wild Escape

- Wireless Locator (RSSI Logic): Designed a treasure-hunt sensor with two ESP32s; used signal strength to adjust the frequency of a buzzer to detect a proximity.
- Multi-Sensor Puzzle: Combined LDR, RFID, and ultrasonic sensors to provide feedback on the game through alerts on the buzzer.
- more.....

SKILLS

IoT & Embedded Systems

- ESP32 Programming
- Arduino (C/C++)
- Sensor calibration (DHT11, MQ135, UV, Rain)
- Circuit assembly & troubleshooting

Cloud & Connectivity

- MQTT (HiveMQ Cloud)
- Node-RED Dashboard & Flow
- ThingSpeak API

Software & Database

- MongoDB
- Python
- HTML / CSS

LANGUAGES

- English – Intermediate
- Chinese – Native
- Malay – Conversational